Data Cleaning & EDA Summary Report

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Cleaning Challenges

The main challenge across both datasets was recognising when unusual values were not errors at all, but genuine business events requiring judgement rather than a blanket rule. In the Online Retail dataset, nearly a quarter of transactions had no CustomerID, too large a share to drop, so these rows were retained and only excluded when customer-level analysis was needed. Extreme price outliers, some as high as £38,970, were traced back to their source and found to be internal entries such as Amazon marketplace fees, postage charges, and bad debt adjustments rather than real sales, and were removed.

Negative quantity values were split between genuine cancellations and true data errors using invoice number patterns. In the Netflix dataset, the main challenge was untangling co-production entries in the country column, where multiple countries were listed together in a single field, which would have distorted any country level analysis if left unaddressed.

Key EDA Findings

Online Retail

- United Kingdom generates the overwhelming majority of revenue at £8.26 million, far ahead of the Netherlands at £283K.
- Monthly revenue climbs steadily from September, peaking at £1.42 million in November, before dropping sharply in December.
- A small group of customers, led by one at £278,778, account for a disproportionate share of total revenue.

Netflix

- Movies outnumber TV Shows by more than 2 to 1, though TV Shows remain a substantial share of the catalogue.
- Content additions grew explosively from 2016 onward, peaking at 2,016 titles added in 2019.
- The catalogue skews toward mature audiences, with TV-MA and TV-14 together accounting for the majority of titles.

Top Insights

1. Both businesses show heavy geographic concentration, Online Retail in the United Kingdom and Netflix in United States produced content, suggesting real opportunity in underrepresented markets.
2. Seasonal and temporal patterns matter, Online Retail's November revenue spike and Netflix's 2016 to 2019 growth surge both point to timing as a key driver of business outcomes.
3. A small number of customers or content categories often account for a disproportionate share of value, seen in Online Retail's top spenders and Netflix's concentration in Dramas and International Movies.
4. Data quality issues are frequently disguised as legitimate records, non-product entries in Online Retail and co-production strings in Netflix both required tracing values back to their source rather than accepting them at face value.